PATENT APPLICATION · PROVISIONAL DRAWINGS

Cross-Botanical Terpene Matching & Multisensory Cannabis Experience Curation System

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PATENT PENDING

FIG. 2 — Terpene-to-audio parameter mapping matrix. Each terpenoid is associated with a primary receptor pathway and neurochemical effect class. These receptor-level properties are translated into Spotify audio feature parameters (tempo, key, mode, energy, valence, acousticness) and instrumental timbre recommendations, enabling cross-modal synchronisation of the olfactory-limbic and auditory experience streams. Eight primary cannabis terpenes are mapped. BPM ranges, key/mode assignments, and Spotify parameter values are derived from receptor pharmacology and neuroacoustic literature.

TERPENOID	RECEPTOR PATHWAY	NEUROCHEMICAL EFFECT	BPM RANGE	KEY / MODE	ENERGY	VALENCE	EO SOURCE (NON-CANNABIS)
β-Caryophyllene $C_{15}H_{24}$ · Sesquiterpene	CB2 / TRPA1	Anti-inflammatory, analgesic, anxiolytic	65–80	MINOR Aeolian / Dorian	0.28	0.35	Black pepper, clove, copaiba, cubeb
d-Limonene $C_{10}H_{16}$ · Monoterpene	5-HT1A / Dopamine	Anxiolytic, mood-elevating, antidepressant	115–130	MAJOR Ionian / Lydian	0.82	0.88	Lemon, sweet orange, bergamot, grapefruit
Myrcene $C_{10}H_{16}$ · Monoterpene	GABA-A / A2a / CB1	Sedating, analgesic, muscle relaxant	55–70	MINOR Phrygian / Aeolian	0.18	0.30	Lemongrass, bay laurel, hops extract
Linalool $C_{10}H_{18}O$ · Monoterpenol	GABA-A / 5-HT1A	Anxiolytic, sedating, anti-stress	75–90	MAJOR Lydian / Mixolydian	0.35	0.72	Lavender, coriander, basil, ho wood
α-Pinene $C_{10}H_{16}$ · Monoterpene	AChE inhibitor / GABA	Alerting, memory enhancement, bronchodilation	100–118	MAJOR Ionian / Mixolydian	0.65	0.62	Pine needle, rosemary, eucalyptus, juniper
α-Humulene $C_{15}H_{24}$ · Sesquiterpene	CB2 / PPARγ	Anti-inflammatory, appetite suppressant	70–85	MINOR Dorian / Aeolian	0.38	0.42	Hops, coriander, sage, clary sage

TERPENOID	RECEPTOR PATHWAY	NEUROCHEMICAL EFFECT	BPM RANGE	KEY / MODE	ENERGY	VALENCE	EO SOURCE (NON-CANNABIS)
Terpinolene $C_{10}H_{16}$ · Monoterpene	SDR / Antioxidant	Uplifting, mildly sedating, antioxidant	118–138	MAJOR Lydian / Ionian	0.76	0.78	Tea tree, nutmeg, cumin, parsnip seed
Ocimene $C_{10}H_{16}$ · Monoterpene	AChE inhibitor / Antiviral	Uplifting, antifungal, decongestant	108–125	MAJOR Ionian / Lydian	0.70	0.80	Basil, parsley, mint, tarragon

Major key — activating / uplifting neurochemical profile (5-HT1A, AChE, Dopamine)

Minor key — grounding / sedating profile (GABA-A, CB2, A2a/CB1)

Methodology note: BPM ranges and Spotify audio parameter values (energy 0–1, valence 0–1) are derived from receptor pharmacology of each terpenoid's primary target pathway. Major/minor key assignment follows the established neuroacoustic correspondence between GABAergic/inhibitory receptor activity and minor-mode perception, and monoaminergic/excitatory activity and major-mode perception. Mode assignments (Aeolian, Dorian, Lydian, etc.) add a secondary affective layer calibrated to the neurochemical effect sub-type (e.g., Dorian for CB2-dominant profiles with moderate warmth; Phrygian for deep GABA-A sedation). EO sources listed are non-cannabis botanical species confirmed to contain the corresponding terpenoid as a primary or significant secondary constituent, consistent with the terpene source-independence principle.